Nonlinear Optimization Notes

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Nonlinear Optimization

Disclaimer
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Colophon
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Introduction

Goals and Motivation

The final goal of this journey is to develop the skills and background necessary to recognize, formulate, and solve optimization problems. A mathematical *optimization problem* has the form

$$\begin{array}{ll} \underset{x \in \mathbf{V}}{\text{minimize}} & f(x) \\ \text{subject to} & x \in \Omega \end{array}$$

where $\mathbf{V}$ is the ambient space of decision variables (we consider general vector spaces), and $\Omega \subseteq \mathbf{V}$ is the *constraint* or *feasible* set, $x \in \mathbf{V}$ is the *optimization variable* of the problem, the function $f : \mathbf{V} \to \mathbf{W}$ is the *objective* function (linear or nonlinear), with $\mathbf{W}$ the codomain. Define $x^\star$ as the *optimal* value or *solution* to the problem stated above.

An example of $\mathbf{V}$ might be $\mathbf{R}^n$.

For many aplications, the codomain $\mathbf{W}$ might be the extended real line $\mathbf{R} \cup \{\infty\}$.

Even when the objective and constraints are "well behaved", solving the optimization problem is surprisingly difficult (with a few exceptions). Approaches to the general problem usually involve some compromise, such as very long computation time, or the possibility of not finding the solution... **In general, optimization problems are unsolvable.**

Less dramatically: in full generality and without structure, global optimization is computationally intractable (typically NP-hard) for most problems.

Families or classes of optimization problems are typically characterized by particular forms of the

- objective function f, and
- constraint set Ω,

which can be labeled as convex (CVX) and non-convex (NCVX). This is summarized in Fig. 1. The most desirable ones are of CVX nature for reasons that will become apparent very soon.

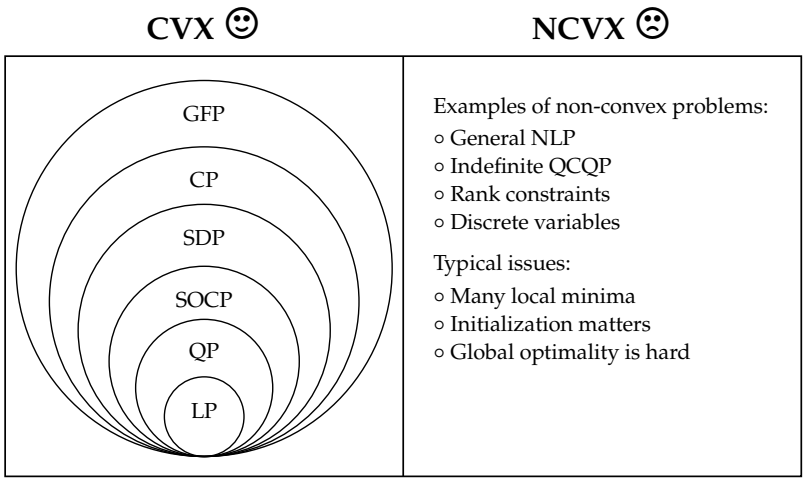

Figure 1: CVX and NCVX optimization problem families (some examples).
– CVX: LP = linear programming, QP = quadratic programming, SOCP = secnd-order cone programming, SDP = semidefinite programming, CP = conic programming, GFP = geometric programming.
– NCVX: NLP = nonlinear programming and QCQP = quadratically constrained quadratic programming (indefinite case).

These notes are organized in three parts much like the course. The first being problem formulation and the foundations of CVX theory, emphasizing the structural properties that make CVX problems tractable and widely useful in modeling. The second covers optimality conditions and duality theory. We then move to methods for NCVX optimization problems.

Mathematical Background

TO BE ADDED

Convex Optimization

1 Convex Sets

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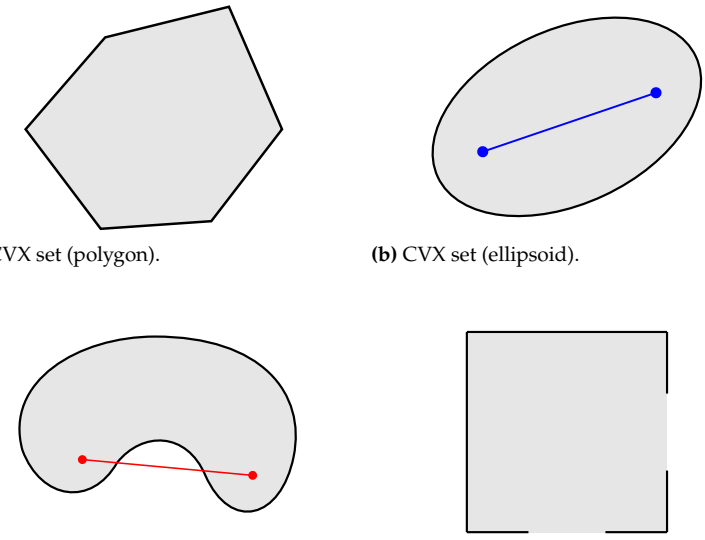
1.1 Definition

Definition 1 (Convex Set) *A set S in a vector space $\mathbf{V}$ is* convex *iff*

$$(1-\alpha)x + \alpha y \in S \tag{1.1}$$

for all $x, y \in S$ and $\alpha \in [0,1]$.

A set is CVX if every point has an unobstructed "line of sight" to every other point, meaning the straight path between them never leaves the set's boundaries. Fig. 1.1 illustrates this intuition behind CVX sets.



(a) CVX set (polygon).

(b) CVX set (ellipsoid).

(c) NCVX set (kidney).

(d) NCVX (square w/o parts of border).

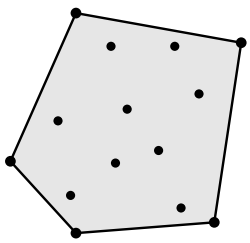
Figure 1.1: Some examples of CVX and NCVX sets. (a) The hexagon (including its boundary, drawn darker) is CVX; (b) The ellipsoid is CVX; (c) The kidney-shaped region is NCVX because the segment connecting the two marked points leaves the set; (d) The square includes only part of its boundary; this incomplete boundary makes the set NCVX.

Definition 2 (Convex Combination) *We call a point in the form $\alpha_1 x_1 + \ldots + \alpha_k x_k$, where $\mathbf{1}^\top \alpha = \sum_k \alpha_k = \alpha_1 + \ldots + \alpha_k = 1$, $\alpha \geq 0$, and $x_1, \ldots, x_k \in S \subset \mathbf{V}$, a* convex combination. *It can also be thought of as a* mixture *or* weighted average, *with a proportion α_i of x_i in it.*

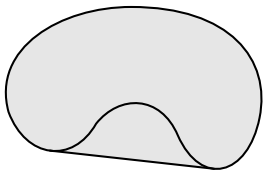
Definition 3 (Convex Hull) *Let S be any subset of a vector space $\mathbf{V}$. We define the* convex hull *of S, denoted by* $\mathrm{conv}(S)$, *to be the set of all finite* convex combinations *of elements of S, i.e.*

$$\mathrm{conv}(S) = \left\{ \sum_{i=1}^{n} \alpha_i x_i \;\middle|\; x_i \in S, \alpha \geq 0, \mathbf{1}^\top \alpha = 1 \right\}. \tag{1.2}$$

Remark 1 Note that $\mathrm{conv}(S)$ is always convex. Generally, $S \subseteq \mathrm{conv}(S)$. Whenever a set S is CVX, it coincides with its convex hull, that is, $\mathrm{conv}(S) = S$. Moreover, $\mathrm{conv}(S)$ is the smallest CVX set containing S.



(a) CVX hull of a polygon.



(b) CVX hull of the kidney shape.

Figure 1.2: Examples of the CVX hull. For the kidney-like shape, we see that the CVX hull contains the NCVX set.

1.2 Simple Convex Sets

1.2.1 Hyperplanes

Definition 4 (Hyperplane) *An* hyperplane *in a vector space* $\mathbf{V}$ *with inner product* $\langle \cdot, \cdot \rangle$ *is a set of the form*

$$H_{s,r} = \{x \in \mathbf{V} \mid \langle s, x \rangle = r\} \tag{1.3}$$

for some $s \in \mathbf{V} \setminus \{0\}$ *and* $r \in \mathbf{R}$.

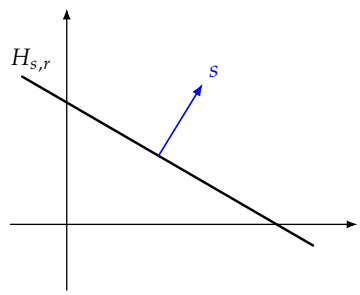


Figure 1.3: Hyperplane with normal s.

1.2.2 Half-Spaces

Definition 5 (Closed Half-Space) *A* closed half-space *in a vector space* $\mathbf{V}$ *with inner product* $\langle \cdot, \cdot \rangle$ *is a set of the form*

$$H^-_{s,r} = \{x \in \mathbf{V} \mid \langle s, x \rangle \leq r\} \tag{1.4}$$

for some $s \in \mathbf{V} \setminus \{0\}$ *and* $r \in \mathbf{R}$.

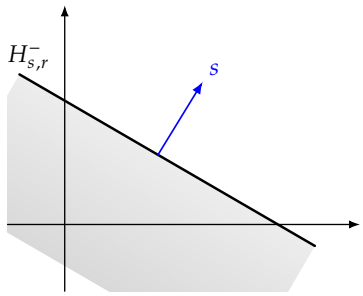


Figure 1.4: Closed half-space defined by a hyperplane; the boundary is included.

1.2.3 Norm Balls

Definition 6 (Norm Ball) *A* norm ball *in a vector space* $\mathbf{V}$ *with norm* $\|\cdot\|_p$ *is a set of the form*

$$B_p(c, r) = \{x \in \mathbf{V} \mid \|x - c\|_p \leq r\} \tag{1.5}$$

for some $c \in \mathbf{V}$ *and* $r \in \mathbf{R}_+$.

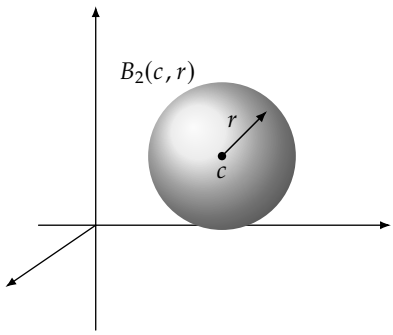


Figure 1.5: Norm ball of radius r centered at c. The plot is a ball living in $\mathbf{R}^3$ with Euclidean norm.

For instance, frequently used norms in $\mathbf{R}^n$ include:

- L^p-norm (or p-norm)

$$\|x\|_p = (|x_1|^p + \ldots + |x_n|^p)^{1/p}$$

- L^1-norm (Manhattan/taxicab norm)

$$\|x\|_1 = |x_1| + \ldots + |x_n|$$

- L^2-norm (Euclidean norm)

$$\|x\|_2 = (|x_1|^2 + \ldots + |x_n|^2)^{1/2}$$

- L^∞-norm (Chebyshev/infinity norm)

$$\|x\|_\infty = \lim_{p \to +\infty} \|x\|_p = \max_n |x_n|$$

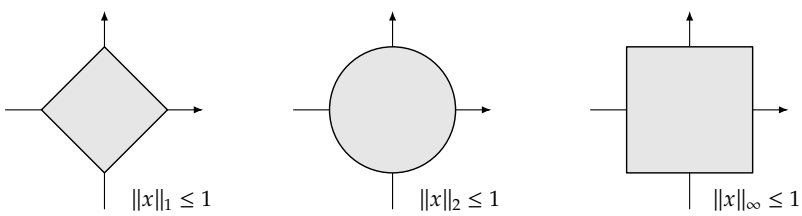


Figure 1.6: Example of norms in 2D.

1.2.4 Cones

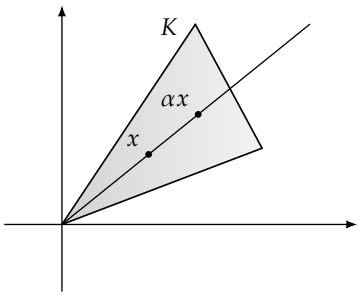


Figure 1.7: Cone containing the origin and closed under nonnegative scalar multiplication.

Definition 7 (Cone) *A set K in a vector space $\mathbf{V}$ is a* convex cone *if K is a convex set and*

$$\mathbf{R}_+ K = \{\alpha x \mid \alpha \in \mathbf{R}_+, x \in K\} \subseteq K. \tag{1.6}$$

Remark 2 Notice we require that cones contain the origin.

Remark 3 Cones are specially important as they can be used to define generalized inequatities (i.e., $\preceq_K, \prec_K, \succeq_K, \succ_K$).

Some examples of a cone include:

- Non-negative orthant in $\mathbf{R}^n$, given by

$$\mathbf{R}^n_+ = \left\{x \in \mathbf{R}^n \mid x_i \geq 0,\ i = 1, \ldots, n\right\}; \tag{1.7}$$

- Lorentz cone or second-order cone in $\mathbf{R}^{n+1}$, given by

$$L^{n+1} = \left\{(x, x_{n+1}) \in \mathbf{R}^n \times \mathbf{R} \mid \|x\|_2 \leq x_{n+1}\right\}; \tag{1.8}$$

- Positive semidefinite cone in $\mathbf{S}^n$, given by

$$\mathbf{S}^n_+ = \left\{X \in \mathbf{S}^n \mid X \succeq 0\right\}. \tag{1.9}$$

  (Eigenvalue decomposition can be useful for this proof.)

1.3 Operations that Preserve Convexity

1.3.1 Intersection of Convex Sets

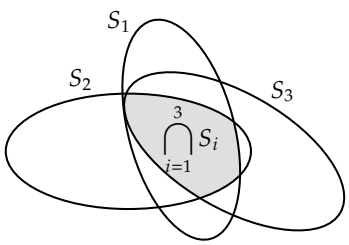


Figure 1.8: Example of an intersection.

Theorem 1 *If $\{S_i : i \in \mathcal{I}\}$ is a family of convex sets in a vector space $\mathbf{V}$, then their intersection $\bigcap_{i \in \mathcal{I}} S_i$ is a convex set. Index set $\mathcal{I}$ may be uncountable.*

Remark 4 Unions of convex sets are generally not convex.

A (convex) *polytope* is a multi-dimensional bounded polyhedron, i.e., a bounded intersection of finitely many closed half-spaces. In $\mathbf{R}^2$ polytopes are (convex) *polygons*, and in $\mathbf{R}^3$ they are (convex) *polyhedra*.

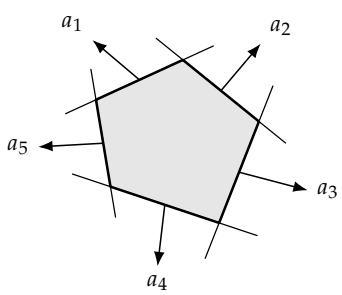


Figure 1.9: Polytope in 2D.

Definition 8 (Polyhedron and polytope) *Let $\mathbf{V}$ be a vector space and $\mathbf{V}^*$ denote its dual space. A (convex)* polyhedron *is a set of the form*

$$P = \left\{x \in \mathbf{V} \,\middle|\, \langle a_i, x\rangle \leq b_i, \langle c_j, x\rangle = d_j\right\}, \tag{1.10}$$

for $i = 1, \ldots, m$ (potentially multiple half-spaces) and $j = 1, \ldots, p$ (potentially multiple hyperplanes), where $a_i, c_j \in \mathbf{V}^*$ and $b \in \mathbf{R}^m$, $d \in \mathbf{R}^p$. A polytope *is a bounded polyhedron.*

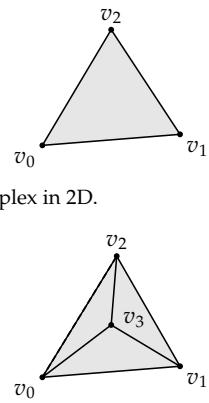
Remark 5 Terminology might vary; some reverse these definitions, defining polytope as the general object and polyhedron as bounded.

Simplexes (or simpleces) are another important family which generalizes the notion of a triangle or tetrahedron to arbitrary dimensions. They are "simplex" because, in n dimensions, they're the simplest polytope.

Definition 9 (Simplex) *Let* $\mathbf{V}$ *be a vector space. Points* $v_0, \ldots, v_n \in \mathbf{V}$ *are* affinely independent *if* $\{v_1 - v_0, \ldots, v_n - v_0\}$ *is linearly independent in* $\mathbf{V}$. *The (convex)* n-simplex *determined by* $v_0, \ldots, v_n$ *is*

$$\Delta^n = \operatorname{conv}(v_0, \ldots, v_n) = \left\{ \sum_{i=0}^{n} \alpha_i v_i \;\middle|\; \alpha \succeq 0, \mathbf{1}^\mathsf{T}\alpha = 1 \right\}. \tag{1.11}$$

Remark 6 The convex hull of a set S is the union of the convex hulls of all finite subsets of S. In particular, it can be viewed as the union of all simplices spanned by points of S.



(a) Simplex in 2D.

(b) Simplex in 3D.

Figure 1.10: Illustration of simpleces.

1.3.2 Transformation by Affine Map

Definition 10 *An affine map* $A : \mathbf{V} \to \mathbf{W}$ *is a map of the form*

$$A(v) = L(v) + w \tag{1.12}$$

for some linear map $L : \mathbf{V} \to \mathbf{W}$ *and some* $w \in \mathbf{W}$.

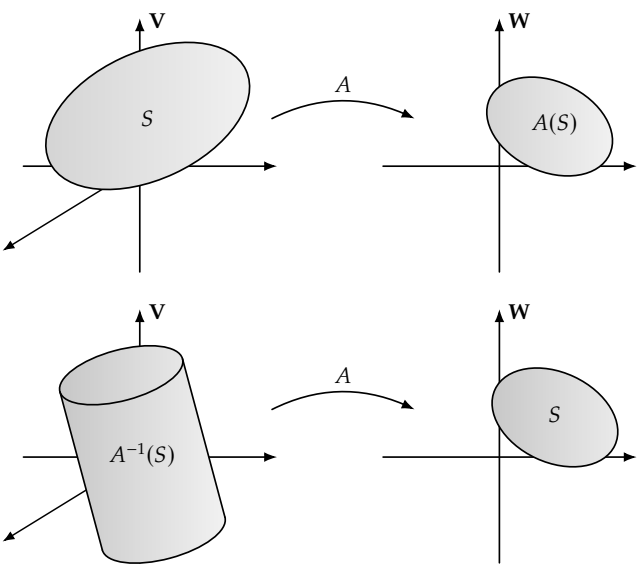


Figure 1.11: (Top) Push-forward by affine map; (Bottom) Pull-back by affine map.

Theorem 2 (Push-Forward by Affine Map) *If* $S \subset \mathbf{V}$ *is a convex set and* $A : \mathbf{V} \to \mathbf{W}$ *an affine map, then*

$$A(S) = \{A(v) \mid v \in S\} \tag{1.13}$$

is a convex set.

Remark 7 *Ellipsoids* are the image of a norm ball under an affine transformation, hence CVX sets as well.

The notation A^{-1} should not be confused with that for inverse function, although it coincides with the one for bijections in that the inverse image of S under A is the image of S under A^{-1}. An example is the projection from a higher dimension to a lower-dimension one.

Theorem 3 (Pull-Back by Affine Map) *If $S \subset \mathbf{W}$ is a convex set and $A : \mathbf{V} \to \mathbf{W}$ an affine map, then the* preimage *or* inverse image

$$A^{-1}(S) = \{v \in \mathbf{V} \mid A(v) \in S\} \tag{1.14}$$

is a convex set.

Remark 8 The affine map A does not need to be invertible.

1.4 Simple Proofs

Example 1 Show that the intersection of a family of convex sets is convex.

Proof. Let $\{S_i : i \in \mathcal{I}\}$ be a family of convex sets, where $\mathcal{I}$ is an arbitrary index set, and consider $S = \bigcap_{i\in\mathcal{I}} S_i$. We show that S is convex.

Let $x, y \in S$ and let $\alpha \in [0, 1]$. Since

$$x, y \in \bigcap_{i\in I} S_i,$$

it follows that

$$x, y \in S_i, \quad \forall i \in \mathcal{I}.$$

Since each S_i is convex,

$$(1-\alpha)x + \alpha y \in S_i, \quad \forall i \in \mathcal{I}.$$

Therefore

$$(1-\alpha)x + \alpha y \in \bigcap_{i\in I} S_i = S.$$

Hence S is convex. □

Example 2 Show that the image of a convex set under an affine map is convex.

Proof. Let $S \subseteq \mathbf{V}$ be a convex set and let $A : \mathbf{V} \to \mathbf{W}$ be an affine map. Now, let $y_1, y_2 \in A(S)$ and let $\alpha \in [0, 1]$.

Then there exist $x_1, x_2 \in S$ such that

$$y_1 = A(x_1), \quad y_2 = A(x_2).$$

Since S is convex,

$$(1-\alpha)x_1 + \alpha x_2 \in S.$$

Moreover, since A is affine,

$$\begin{aligned}(1-\alpha)y_1 + \alpha y_2 &= (1-\alpha)A(x_1) + \alpha A(x_2)\\ &= A\big((1-\alpha)x_1 + \alpha x_2\big).\end{aligned}$$

Hence

$$(1-\alpha)y_1 + \alpha y_2 \in A(S).$$

Therefore $A(S)$ is convex. □

Example 3 Show that if S_1 and S_2 are convex sets, then their cartesian product $S_1 \times S_2$ is convex.

Proof. Let $(x_1, x_2), (y_1, y_2) \in S_1 \times S_2$ and let $\alpha \in [0, 1]$. Then

$$(1-\alpha)(x_1, x_2) + \alpha(y_1, y_2) = \big((1-\alpha)x_1 + \alpha y_1, (1-\alpha)x_2 + \alpha y_2\big).$$

Since S_1 and S_2 are convex,

$$(1-\alpha)x_1 + \alpha y_1 \in S_1, \qquad (1-\alpha)x_2 + \alpha y_2 \in S_2.$$

Therefore

$$(1-\alpha)(x_1, x_2) + \alpha(y_1, y_2) \in S_1 \times S_2.$$

Hence $S_1 \times S_2$ is convex. □

Example 4 Show that if S_1 and S_2 are convex sets in a vector space $\mathbf{V}$, then

$$S_1 \pm S_2 = \{x_1 \pm x_2 \mid x_1 \in S_1,\ x_2 \in S_2\}$$

is convex.

Proof. Since S_1 and S_2 are convex, their cartesian product $S_1 \times S_2$ is convex. Consider the affine map

$$A : \mathbf{V} \times \mathbf{V} \to \mathbf{V}, \quad A(x_1, x_2) = x_1 \pm x_2.$$

Then

$$A(S_1 \times S_2) = \{x_1 \pm x_2 \mid x_1 \in S_1,\ x_2 \in S_2\} = S_1 \pm S_2.$$

Since the image of a convex set under an affine map is convex, it follows that $S_1 \pm S_2$ is convex. □

2 Convex Functions

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2.1 Definition

Definition 11 (Convex Function) *A function $f : \mathbf{V} \to \mathbf{R} \cup \{+\infty\}$ in a vector space $\mathbf{V}$ is convex if $f \not\equiv \infty$ and*

$$f\big((1-\alpha)x + \alpha y\big) \leq (1-\alpha)f(x) + \alpha f(y) \tag{2.1}$$

for all $x, y \in S \subset \mathbf{V}$ and $\alpha \in [0,1]$.

Remark 9 Take the convention $0(+\infty) := 0$.

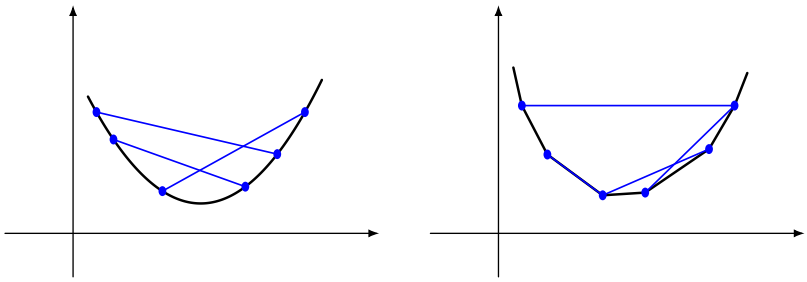

Figure 2.1: Example of CVX functions. The functions does not need to be smooth as long as it satisfies the definition of a CVX function.

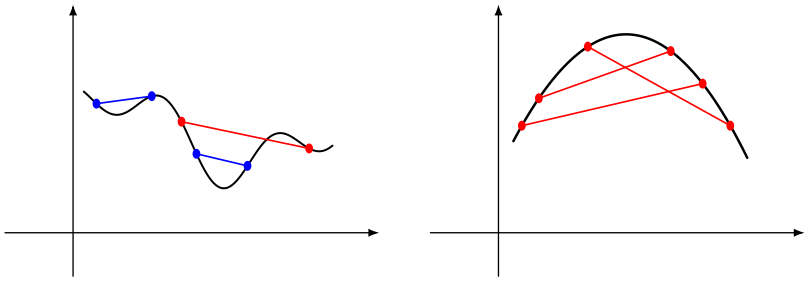

Figure 2.2: Example of NCVX functions.

Definition 12 *The domain of f is the set* $\operatorname{dom} f = \{x \in \mathbf{V} \mid f(x) < +\infty\}$.

Theorem 4 *If f is convex, its domain is convex.*

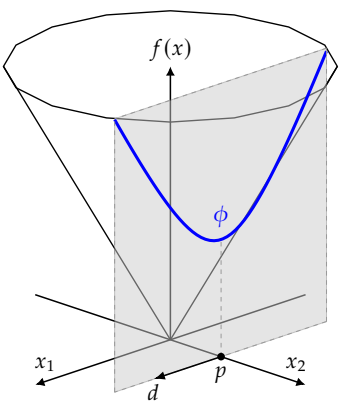

Figure 2.3: Illustration of the 1D CVX property for a cone shape living in $\mathbf{R}^3$.

2.1.1 Other Convexity Criteria

Verifying that a function is CVX by definition becomes nontrivial as soon as the function is nonlinear. To this end, extra criteria are considered.

Proposition 5 (Convexity is a 1D property) *A function f is convex iff $\phi : \mathbf{R} \to \mathbf{R} \cup \{+\infty\}$,*

$$\phi(t) = f(p + td), \tag{2.2}$$

is convex for any p and d in $\mathbf{V}$.

Theorem 6 (First-Order Criterion) *Let S be an open subset of $\mathbf{R}^n$ and $f : S \to \mathbf{R}$ be a differentiable function. Then,*

$$f \text{ is convex} \iff f(x) \geq f(y) + [\nabla f(y)]^{\mathsf{T}}(x - y) \tag{2.3}$$

for all $x, y \in S$.

Remark 10 The first-order condition is mostly useful in the $\Longleftarrow$ "direction" and shows that every convex function admits global affine lower bounds. The tangent hyperplane at any point $y \in S$ *underestimates the function* everywhere. This property is frequently *used to derive inequalities*.

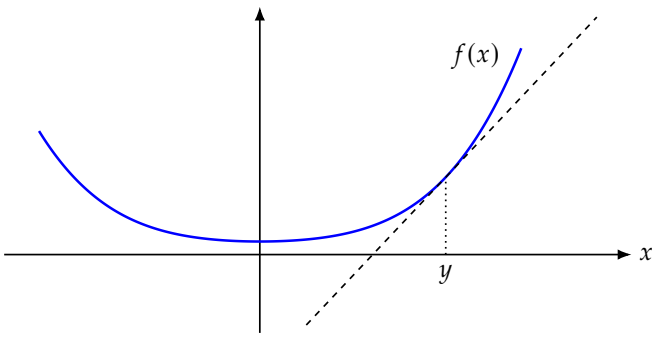

Figure 2.4: First-order characterization of convexity. A convex function lies above its first-order affine approximation.

Theorem 7 (Second-Order Criterion) *Let S be an open subset of $\mathbf{R}^n$ and $f : S \to \mathbf{R}$ be a twice-differentiable function. Then,*

$$f \text{ is convex} \iff \nabla^2 f(x) \succeq 0 \tag{2.4}$$

for all $x \in S$.

Remark 11 The second-order characterization is mostly useful in the $\Longrightarrow$ "direction" as it provides a *practical criterion for proving* that f is convex. Contrary to the first-order criterion, in general, the second-order Taylor expansion is *not an upper bound* on $f(x)$.

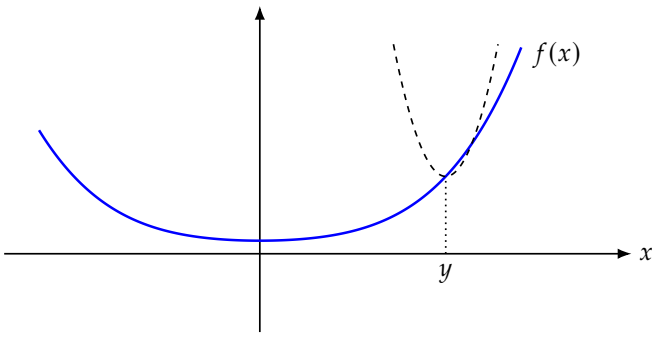

Figure 2.5: Second-order condition for convexity. Positive semidefiniteness of curvatuve ensures the function remains above its quadratic approximation.

Theorem 8 *For any $A \in \mathbf{S}^n_{++}$ and $B \in \mathbf{S}^n$ there exists $S \in \mathbf{R}^{n\times n}$ such that*

$$A = SS^T \text{ and } B = S\Lambda S^T \tag{2.5}$$

where $\Lambda \in \mathbf{R}^{n\times n}$ is diagonal, with the eigenvalues of $A^{-1/2}BA^{-1/2}$.

2.2 Simple Convex Functions

2.2.1 Affine Functions

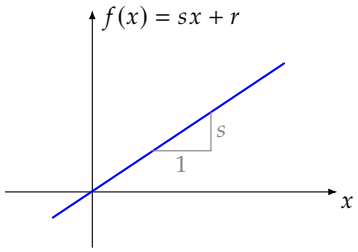

Figure 2.6: An affine function in **R**.

Definition 13 (Affine Function) *An affine function $f : \mathbf{V} \to \mathbf{R}$ is a map of the form*

$$f(v) = l(v) + r \tag{2.6}$$

for some linear function $l : \mathbf{V} \to \mathbf{R}$ and some $r \in \mathbf{R}$.

Theorem 9 *An affine function is convex.*

2.2.2 Norms

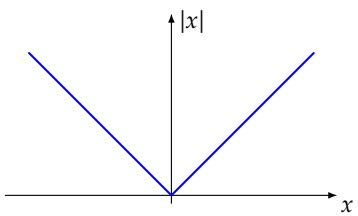

Figure 2.7: Illustration of a norm in 2D.

Definition 14 (Norm) *Let $\mathbf{V}$ be a vector space. A function $\|\cdot\| : \mathbf{V} \to \mathbf{R}$ is called a norm if, for all $x, y \in V$ and all $\alpha \in \mathbf{K}$ (i.e., $\mathbf{R}$ or $\mathbf{C}$), it satisfies:*

1. *Non-negativity: $\|x\| \geq 0$,*
2. *Positiveness: $\|x\| = 0$ iff $x = 0$,*
3. *Homogeneity: $\|\alpha x\| = |\alpha|\,\|x\|$,*
4. *Subadditivity: $\|x + y\| \leq \|x\| + \|y\|$.*

Theorem 10 *A norm is a convex function.*

2.2.3 Indicators

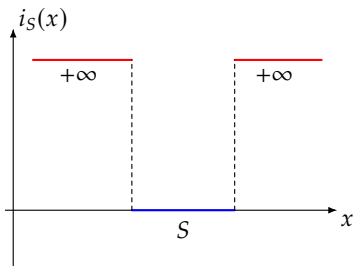

Figure 2.8: Illustration of the indicator function.

Definition 15 *The indicator of a set $S \subset \mathbf{V}$ is the function*

$$i_S : \mathbf{V} \to \mathbf{R} \cup \{+\infty\}, \quad i_S(x) = \begin{cases} 0, & \textit{if } x \in S; \\ +\infty & \textit{otherwise.} \end{cases} \tag{2.7}$$

Remark 12 We can think of this indicator function as an ideal penalty function. Whenever a point is outside the set of interest, it pays a hefty price (explodes to $+\infty$).

Theorem 11 *The indicator of a convex set is a convex function.*

2.3 Operations that Preserve Convexity

2.3.1 Conic Combination

Theorem 12 (Conic Combination) *Let $f_i : \mathbf{V} \to \mathbf{R} \cup \{+\infty\}$ be convex functions and $\alpha_i \geq 0$ for $i = 1, \ldots, n$. If $\bigcap_i \operatorname{dom} f_i \neq \emptyset$, then*

$$f = \alpha_1 f_1 + \ldots + \alpha_n f_n \tag{2.8}$$

is convex.

2.3.2 Composition with Affine Map

Theorem 13 (Composition with Affine Map) *Let $A : \mathbf{V} \to \mathbf{W}$ be an affine map, $f : \mathbf{W} \to \mathbf{R} \cup \{+\infty\}$ be convex, and $A(\mathbf{V}) \cap \operatorname{dom} f \neq \emptyset$. Then*

$$f \circ A : \mathbf{V} \to \mathbf{R} \cup \{+\infty\} \tag{2.9}$$

is convex.

2.3.3 Pointwise Supremum

Theorem 14 (Pointwise Supremum) *Let $f_i : \mathbf{V} \to \mathbf{R} \cup \{+\infty\}$, $i \in \mathcal{I}$, be a family of convex functions and suppose that $f = \sup_{i \in \mathcal{I}} f_i \not\equiv +\infty$. Then, $f : \mathbf{V} \to \mathbf{R} \cup \{+\infty\}$ is convex.*

2.4 Simple Proofs

Example 5 Show that the function

$$f : \mathbf{S}^n_{++} \to \mathbf{R}, \qquad f(X) = \operatorname{tr}(X^{-1})$$

is convex.

Proof. TODO. □

Example 6 Show that the function

$$f : \mathbf{S}^n_{++} \to \mathbf{R}, \qquad f(X) = -\log\det(X)$$

is convex.

Proof. TODO. □

Main References

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